NEET UG Chemistry Tutorial

Comprehensive Masterclass & Assessment: Aldehydes, Ketones, and Carboxylic Acids

1. Syllabus Orientation & Learning Objectives

The carbonyl group ($>\mathrm{C}=\mathrm{O}$) and the carboxyl group ($-\mathrm{COOH}$) constitute the core of organic oxygen chemistry in the NEET UG syllabus. Every year, 3 to 4 direct questions test named preparations, nucleophilic addition-elimination mechanisms, relative acidities, and diagnostic tests (Tollens', Fehling's, and Iodoform tests).

Key Competencies for NEET:

- Deduce IUPAC and common names of aliphatic and aromatic carbonyls and carboxylic acids.
- Analyze the electronic structure, polarity, and steric profile of the carbonyl carbon.
- Master specific industrial and laboratory preparations: Rosenmund, Stephen, Etard, Gatterman-Koch, and Friedel-Crafts acylation.
- Formulate mechanistic pathways for nucleophilic addition with HCN, $\mathrm{NaHSO_3}$, alcohols, and nitrogen derivatives ($\mathrm{H_2N-Z}$).
- Predict cross-aldol condensation products, Cannizzaro disproportionations, and decarboxylation routes.
- Correlate substituent inductive ($\pm I$) and resonance ($\pm R$) effects with carboxylic acid dissociation constants (pK_a).

2. Structure and Polarity of the Carbonyl Group

The carbonyl carbon is sp^2-hybridised and forms three σ-bonds oriented in a single plane with approximately 120° bond angles (trigonal coplanar geometry). The unhybridised $2p$-orbital of carbon overlaps laterally with a parallel $2p$-orbital of the oxygen atom to construct a π-bond located above and below the molecular plane.

Oxygen is significantly more electronegative than carbon ($\chi_\mathrm{O} \approx 3.5$ vs. $\chi_\mathrm{C} \approx 2.5$). Consequently, the π-electron density is strongly shifted toward oxygen, inducing an electrophilic (Lewis acidic) centre at the carbon atom (δ^+) and a nucleophilic (Lewis basic) centre at the oxygen atom (δ^-). Carbonyl compounds possess large permanent dipole moments (typically $2.3\text{--}2.8\ \mathrm{D}$), making them distinctly more polar than ethers or aliphatic halides.



Tutorial Diagram 1: Planar trigonal geometry, π-orbital sideways overlap, and dipolar canonical contributors of the carbonyl moiety.

3. Systematic Nomenclature & IUPAC Conventions

In IUPAC nomenclature, the acyclic aliphatic aldehyde name replaces the terminal *-e* of the alkane with *-al*. The aldehydic carbon is designated as carbon-1 ($\mathrm{C}\text{-}1$) by default. For acyclic ketones, the suffix is *-one*, and the chain is numbered to give the carbonyl carbon the lowest possible locant.

- **Ring Carbaldehydes:** When the $-\mathrm{CHO}$ group is directly bonded to a saturated or aromatic ring, the suffix **carbaldehyde** is appended to the full name of the cyclic hydrocarbon (e.g., cyclohexanecarbaldehyde, benzenecarbaldehyde).
- **Polycarbaldehydes:** When three or more aldehydic groups are attached to an unbranched carbon chain, the carbon chain is named without counting the carbonyl carbons in the principal chain, and the suffix *tricarbaldehyde* is applied. Example: $\mathrm{OHC-CH_2-CH(CHO)-CH_2-CHO}$ is **propane-1,2,3-tricarbaldehyde**.
- **Substituent Precedence:** When an aldehyde and a ketone coexist in the same molecule, the aldehyde has higher IUPAC seniority; the ketone oxygen is treated as an **oxo-** substituent. For example, $\mathrm{CH_3-CO-CH_2-CH_2-CHO}$ is **4-oxopentanal**.

Chemical Structure	Common Name	IUPAC Name	NEET Exam Note
HCHO	Formaldehyde	Methanal	Gas; 40% aqueous solution is formalin
$\mathrm{CH_3CHO}$	Acetaldehyde	Ethanal	Volatile liquid (b.p. 294 K)
$\mathrm{CH_2=CH-CHO}$	Acrolein	Prop-2-enal	Unsaturated aldehyde; formed by dehydrating glycerol
$\mathrm{(CH_3)_2C=CH-CO-CH_3}$	Mesityl oxide	4-Methylpent-3-en-2-one	Aldol condensation product of propanone

BOSONANCE

Chemical Structure	Common Name	IUPAC Name	NEET Exam Note
$C_6H_4(CHO)_2$ (1,2-isomer)	Phthaldehyde	Benzene-1,2-dicarbaldehyde	Aromatic dialdehyde
$C_6H_5-COCH_3$	Acetophenone	1-Phenylethan-1-one	Aryl methyl ketone; positive iodoform test
$C_6H_5-CO-C_6H_5$	Benzophenone	Diphenylmethanone	Diaryl ketone; negative iodoform and Fehling's tests

4. Methods of Preparation of Aldehydes and Ketones

4.1 Common Routes (Both Aldehydes and Ketones)

1. **Controlled Oxidation of Alcohols:** Primary alcohols oxidise selectively to aldehydes using Pyridinium Chlorochromate (PCC) in anhydrous CH_2Cl_2. Secondary alcohols oxidise to ketones using chromic anhydride (CrO_3) or acidic dichromate:

$$R-CH_2OH \xrightarrow{PCC,\ CH_2Cl_2} R-CHO$$
$$R-CH(OH)-R' \xrightarrow{CrO_3} R-CO-R'$$

2. **Catalytic Dehydrogenation of Alcohol Vapours:** Passing alcohol vapours over heated heavy metal catalysts (finely divided Cu or Ag at $573\ K$):

$$R-CH_2OH \xrightarrow{Cu,\ 573\ K} R-CHO + H_2$$
$$R-CH(OH)-R' \xrightarrow{Cu,\ 573\ K} R-CO-R' + H_2$$

3. **Reductive Ozonolysis of Alkenes:** Alkenes cleave upon treatment with ozone followed by zinc dust and water (Zn/H_2O). Zinc prevents further oxidation of aldehydes to carboxylic acids by destroying hydrogen peroxide (H_2O_2). Terminal $=CH_2$ yields methanal.

4. **Hydration of Alkynes (Kucherov Reaction):** Addition of water across an alkyne in the presence of dilute H_2SO_4 and mercuric sulphate catalyst ($HgSO_4$) at $333\ K$. Ethyne alone gives ethanal; all higher terminal and internal alkynes give ketones via keto-enol tautomerisation:

$$HC\equiv CH + H_2O \xrightarrow[H_2SO_4]{HgSO_4} [CH_2=CH-OH] \rightleftharpoons CH_3-CHO$$
$$CH_3-C\equiv CH + H_2O \xrightarrow[H_2SO_4]{HgSO_4} [CH_3-C(OH)=CH_2] \rightleftharpoons CH_3-CO-CH_3$$

4.2 Specific Synthesis of Aldehydes

- **Rosenmund Reduction:** Hydrogenation of acyl chlorides over palladium supported on barium sulphate ($Pd/BaSO_4$), partially poisoned with sulfur or quinoline to prevent over-reduction to primary alcohols:

$$R-COCl + H_2 \xrightarrow{Pd\text{-}BaSO_4} R-CHO + HCl$$

NEET Point: Formaldehyde cannot be prepared by this method because formyl chloride ($HCOCl$) is unstable at room temperature.

- **Stephen Reaction:** Reduction of alkyl or aryl nitriles with stannous chloride and hydrochloric acid ($SnCl_2 + HCl$), generating an aldimine hydrochloride salt, followed by warm aqueous hydrolysis:

$$R-C\equiv N + SnCl_2 + 2HCl \longrightarrow R-CH=NH\cdot HCl \xrightarrow{H_3O^+} R-CHO + NH_4Cl$$

- **Selective Reduction with DIBAL-H:** Diisobutylaluminium hydride, $[(CH_3)_2CHCH_2]_2AlH$, selectively reduces nitriles and esters to aldehydes at $195\ K$ without reducing carbon-carbon double or triple bonds:

$$CH_3-CH=CH-CH_2CH_2-CN \xrightarrow[2.\ H_2O]{1.\ DIBAL\text{-}H,\ 195\ K} CH_3-CH=CH-CH_2CH_2-CHO$$

- **Etard Reaction:** Toluene is oxidised by chromyl chloride (CrO_2Cl_2) dissolved in carbon disulfide (CS_2) or CCl_4 to precipitate a brown chromium complex, which yields benzaldehyde upon aqueous hydrolysis:

$$C_6H_5-CH_3 + 2CrO_2Cl_2 \xrightarrow{CS_2} C_6H_5-CH(OCrOHCl_2)_2 \xrightarrow{H_3O^+} C_6H_5-CHO$$

- **Chromic Oxide in Acetic Anhydride:** Oxidation of toluene with CrO_3 in $(CH_3CO)_2O$ at $273\text{--}283\ K$ captures the aldehyde stage as a gem-diacetate intermediate (benzylidene diacetate), preventing over-oxidation. Hydrolysis with dilute acid liberates benzaldehyde:

$$C_6H_5-CH_3 \xrightarrow[273\text{--}283\ K]{CrO_3,\ (CH_3CO)_2O} C_6H_5-CH(OCOCH_3)_2 \xrightarrow[\Delta]{H_3O^+} C_6H_5-CHO + 2CH_3COOH$$

- **Side-Chain Chlorination:** Free-radical side-chain chlorination of toluene yields benzal chloride, which on industrial hydrolysis at $373\ K$ affords benzaldehyde:

$$C_6H_5-CH_3 \xrightarrow{Cl_2,\ h\nu} C_6H_5-CHCl_2 \xrightarrow{H_2O,\ 373\ K} C_6H_5-CHO + 2HCl$$

- **Gatterman-Koch Reaction:** Direct formylation of benzene or alkylbenzenes using carbon monoxide (CO) and dry HCl in the presence of anhydrous $AlCl_3$ and cuprous chloride ($CuCl$):

$$C_6H_6 + CO + HCl \xrightarrow{\text{anhydrous } AlCl_3/CuCl} C_6H_5-CHO + HCl$$

4.3 Specific Synthesis of Ketones

- **From Acyl Chlorides and Dialkylcadmium:** Grignard reagents react with anhydrous cadmium chloride to form dialkylcadmium. Because R_2Cd is less nucleophilic than $RMgX$, it displaces the chloride of acyl chlorides cleanly without attacking the resulting ketone:

$$2R-MgX + CdCl_2 \longrightarrow R_2Cd + 2Mg(X)Cl$$
$$2R'-COCl + R_2Cd \longrightarrow 2R'-CO-R + CdCl_2$$

- **From Nitriles via Grignard Addition:** Nucleophilic addition of a Grignard reagent to a nitrile in anhydrous ether yields an imine salt intermediate, which produces a ketone on acid hydrolysis:

$$CH_3CH_2-C\equiv N + C_6H_5MgBr \xrightarrow{\text{ether}} CH_3CH_2-C(C_6H_5)=NMgBr \xrightarrow{H_3O^+} CH_3CH_2-CO-C_6H_5$$

BOSONANCE

- **Friedel-Crafts Acylation:** Treatment of benzene or substituted arenes with an acyl chloride or acid anhydride in the presence of anhydrous $\mathrm{AlCl_3}$ produces aryl ketones:

$$\mathrm{C_6H_6 + CH_3COCl \xrightarrow{\text{anhydrous } AlCl_3} C_6H_5 - COCH_3 + HCl}$$

5. Physical Properties & Intermolecular Forces

Boiling Point Trend: For compounds of comparable molecular mass (~ 58--$60\,\mathrm{g\,mol^{-1}}$), the boiling point hierarchy reflects the strength of intermolecular forces:

> **Hydrocarbon (non-polar van der Waals) < Ether (weak dipole) < Aldehyde < Ketone (dipole-dipole) < Alcohol (H-bonding) < Carboxylic Acid (extensive H-bonded dimer)**

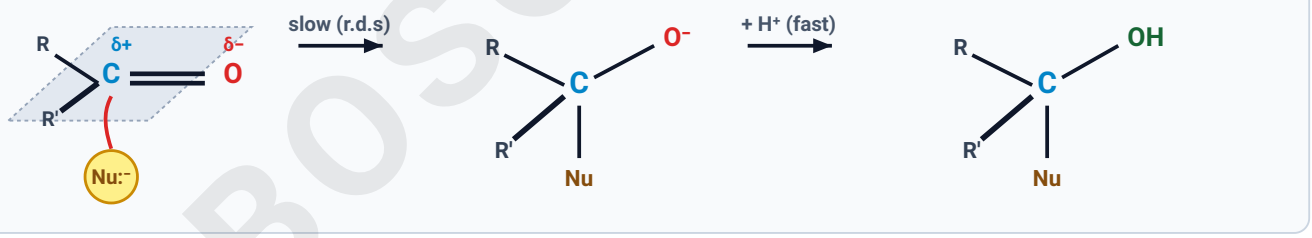
Compound	Structure	Molecular Mass ($\mathrm{g\,mol^{-1}}$)	Boiling Point (K)	Intermolecular Interaction Type
n-Butane	$\mathrm{CH_3(CH_2)_2CH_3}$	58	273	Weak London dispersion forces
Methoxyethane	$\mathrm{CH_3 - O - C_2H_5}$	60	281	Weak dipole-dipole attractions
Propanal	$\mathrm{CH_3CH_2CHO}$	58	322	Permanent dipole-dipole interaction
Acetone	$\mathrm{CH_3COCH_3}$	58	329	Stronger permanent dipole-dipole
Propan-1-ol	$\mathrm{CH_3CH_2CH_2OH}$	60	370	Intermolecular hydrogen bonding
Ethanoic acid	$\mathrm{CH_3COOH}$	60	391	Stable cyclic intermolecular H-bonded dimer

Solubility in Water: Lower aldehydes and ketones (methanal, ethanal, propanone) are miscible with water in all proportions because their carbonyl oxygen acts as an effective hydrogen-bond acceptor with water molecules ($> \mathrm{C=O^{\delta-} \cdots H^{\delta+} - O - H}$). Solubility decreases sharply beyond four carbons as the hydrophobic alkyl chain increases.

6. Chemical Reactions of Aldehydes and Ketones

6.1 Mechanism of Nucleophilic Addition

The carbonyl carbon has planar sp^2 geometry. A nucleophile ($\mathrm{Nu^-}$ or $\mathrm{:Nu-H}$) attacks the electrophilic carbon from a direction approximately perpendicular ($\approx 107^\circ$, Bürgi-Dunitz trajectory) to the plane of the carbonyl group. In this slow, rate-determining step, carbon hybridisation changes from sp^2 to sp^3, producing a tetrahedral alkoxide intermediate. Rapid protonation from the solvent yields the neutral addition product.



Tutorial Diagram 2: Two-step mechanism of nucleophilic addition to the carbonyl carbon via an alkoxide intermediate.

Reactivity Order in Nucleophilic Addition:
$\mathrm{HCHO > CH_3CHO > CH_3CH_2CHO > CH_3COCH_3 > CH_3COCH_2CH_3 > C_6H_5CHO > C_6H_5COCH_3 > C_6H_5COC_6H_5}$
Underlying Causes:

1. **Steric Hindrance:** Ketones possess two bulky alkyl groups flanking the carbonyl carbon, hindering nucleophilic attack and creating steric congestion as bond angles compress from 120° to 109.5°.
2. **Electronic Factor:** Alkyl groups donate electrons inductively ($+I$), dispersing the δ^+ charge and reducing electrophilicity.
3. **Aromatic Resonance Deactivation:** In benzaldehyde, the π-electrons of the benzene ring delocalise into the carbonyl group by resonance, significantly decreasing the positive charge on the carbonyl carbon compared to aliphatic aldehydes.

6.2 Key Nucleophilic Addition Reactions

- **Addition of Hydrogen Cyanide (HCN):** Pure HCN reacts sluggishly because it is a weak acid. Addition is catalysed by base (dilute $\mathrm{OH^-}$), which generates cyanide ion ($\mathrm{CN^-}$), a potent nucleophile:

$$\mathrm{HCN + OH^- \rightleftharpoons :CN^- + H_2O}$$

$$\mathrm{> C=O + CN^- \rightleftharpoons > C(O^-)(CN) \xrightarrow{H^+} > C(OH)(CN) \quad (Cyanohydrin)}$$

Hydrolysis of cyanohydrins yields α-hydroxy acids ($\mathrm{R-CH(OH)-COOH}$).

BOSONANCE

- **Addition of Sodium Hydrogensulphite ($NaHSO_3$):** Aldehydes and methyl ketones form crystalline bisulphite addition products. The nucleophilic sulfur atom attacks the carbonyl carbon:

$$>C=O + NaHSO_3 \rightleftharpoons >C(ONa)(SO_3H) \xrightarrow{\text{proton transfer}} >C(OH)(SO_3Na)\downarrow$$

The equilibrium position favors the adduct for aldehydes, but lies largely to the left for sterically hindered ketones. Because the crystalline product regenerates the original carbonyl upon treatment with dilute mineral acid or base, this reaction is widely used for separation and purification.

- **Addition of Alcohols:** Aldehydes react with one equivalent of anhydrous monohydric alcohol in the presence of dry HCl gas to form a **hemiacetal** (alkoxyalcohol), which reacts with a second equivalent to yield a gem-dialkoxy compound called an **acetal**:

$$R-CHO + R'OH \xrightleftharpoons{\text{dry HCl}} R-CH(OH)(OR') \xrightleftharpoons[-H_2O]{R'OH,\ H^+} R-CH(OR')_2$$

Ketones react with 1,2-diols (ethylene glycol) in dry HCl to form cyclic **ethylene glycol ketals**. These protecting groups are hydrolysed back to parent carbonyls using aqueous mineral acids.

- **Addition of Ammonia Derivatives (H_2N-Z):** Weakly basic nitrogen nucleophiles attack the carbonyl group in an acid-catalysed, reversible addition-elimination sequence:

$$>C=O + H_2N-Z \xrightleftharpoons{H^+} [>C(OH)-NHZ] \xrightarrow{-H_2O} >C=N-Z$$

Group ($-Z$)	Reagent Name	Product Functional Group	Class Name & Diagnostic Property
$-H$	Ammonia (NH_3)	$>C=NH$	Imine
$-R$	Primary amine (RNH_2)	$>C=NR$	Schiff's base (substituted imine)
$-OH$	Hydroxylamine (NH_2OH)	$>C=N-OH$	Oxime (shows geometrical isomerism)
$-NH_2$	Hydrazine (NH_2NH_2)	$>C=N-NH_2$	Hydrazone
$-NHC_6H_5$	Phenylhydrazine	$>C=N-NHC_6H_5$	Phenylhydrazone
$-NH-C_6H_3(NO_2)_2$	2,4-Dinitrophenylhydrazine (Brady's Reagent)	$>C=N-NH-C_6H_3(NO_2)_2$	2,4-DNP derivative; sharp-melting **yellow, orange, or red solid** used to confirm carbonyl group
$-NHCONH_2$	Semicarbazide	$>C=N-NHCONH_2$	Semicarbazone

NEET High-Yield Insight (Semicarbazide): Semicarbazide has the formula $H_2N^1-NH-CO-N^2H_2$. Only the N^1 hydrazine amino group acts as a nucleophile. The N^2 amide amino group's lone pair is strongly delocalised into the adjacent carbonyl group ($-CO-$) by resonance:

$$H_2N-NH-\overset{\overset{O}{\|}}{C}-\ddot{N}H_2 \longleftrightarrow H_2N-NH-\overset{\overset{O^-}{|}}{C}=\overset{+}{N}H_2$$

Thus, N^2 is non-nucleophilic and cannot attack the aldehyde or ketone.

6.3 Reduction Reactions

1. **Reduction to Hydrocarbons:**
   - **Clemmensen Reduction:** The carbonyl group is reduced to a methylene group ($-CH_2-$) using amalgamated zinc and concentrated hydrochloric acid:

$$>C=O + 4[H] \xrightarrow{\text{Zn-Hg / conc. HCl}} >CH_2 + H_2O$$

   Limitation: Unsuitable for substrates containing acid-sensitive groups (such as acetals or tertiary alcohols).
   - **Wolff-Kishner Reduction:** The carbonyl compound is heated with hydrazine (NH_2NH_2) to form a hydrazone, which is then heated with a strong base (KOH) in a high-boiling solvent like ethylene glycol (~ 453--$473\ K$):

$$>C=O \xrightarrow[-H_2O]{NH_2NH_2} >C=NNH_2 \xrightarrow[\text{ethylene glycol, }\Delta]{KOH} >CH_2 + N_2\uparrow$$

   Limitation: Unsuitable for substrates containing base-sensitive functional groups (such as alkyl halides or esters).

6.4 Oxidation and Distinguishing Tests

Aldehydes oxidise readily to carboxylic acids even with mild reagents because they contain a oxidisable hydrogen atom directly bonded to the carbonyl carbon. In contrast, ketones lack this hydrogen and require vigorous conditions (hot concentrated HNO_3 or $KMnO_4$), which cleave carbon-carbon bonds and produce complex mixtures of smaller carboxylic acids (Popoff's rule).

Diagnostic Test	Reagent Composition	Positive Observation	Substrates Giving Positive Result
Tollens' Test (Silver Mirror Test)	Freshly prepared ammoniacal silver nitrate solution: $[Ag(NH_3)_2]^+OH^-$	Bright metallic silver mirror or black silver precipitate: $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$	All aliphatic aldehydes, aromatic aldehydes, and formic acid. Ketones do not react.
Fehling's Test	Equal parts of Fehling A (aqueous $CuSO_4$) and Fehling B (alkaline sodium potassium tartrate / Rochelle salt)	Red-brown precipitate of cuprous oxide (Cu_2O): $RCHO + 2Cu^{2+} + 5OH^- \longrightarrow RCOO^- + Cu_2O\downarrow + 3H_2O$	**Aliphatic aldehydes only.** Aromatic aldehydes (e.g., benzaldehyde) and ketones do not reduce Fehling's solution.

BOSONANCE

Diagnostic Test	Reagent Composition	Positive Observation	Substrates Giving Positive Result
Iodoform Test (Haloform Reaction)	Iodine and aqueous sodium hydroxide ($\mathrm{I_2 + NaOH}$ or NaOI)	Bright yellow crystalline precipitate of iodoform ($\mathrm{CHI_3}$, m.p. 392 K) with characteristic antiseptic odour: $$\mathrm{R-CO-CH_3 + 3NaOI \longrightarrow R-COONa + CHI_3\downarrow + 2NaOH}$$	Compounds containing the acetyl group ($\mathrm{CH_3-C=O}$) or those oxidisable to it ($\mathrm{CH_3-CH(OH)-}$). This includes ethanal, all methyl ketones, ethanol, and secondary methyl carbinols.

6.5 Reactions Governed by α-Hydrogen Acidity

The α-hydrogens of aldehydes and ketones are weakly acidic ($pK_a \approx 19$--20) due to the strong electron-withdrawing inductive effect ($-I$) of the carbonyl group and resonance delocalisation of the resulting conjugate base (enolate anion):

$$\mathrm{-CH_2-C(=O)- + :B \rightleftharpoons H:B^+ + \left[-CH^- - C(=O)- \longleftrightarrow -CH=C(O^-)-\right]}$$

1. Aldol Reaction and Condensation

Carbonyl compounds having at least one α-hydrogen react in dilute alkali (NaOH, $\mathrm{Ba(OH)_2}$) to yield β-hydroxy aldehydes (aldols) or β-hydroxy ketones (ketols). Upon heating, these readily dehydrate via an $\mathrm{E1cB}$ pathway to form conjugated α,β-unsaturated carbonyls:

$$\mathrm{2CH_3CHO \xrightarrow{dil.\ NaOH} CH_3-CH(OH)-CH_2-CHO \xrightarrow[-H_2O]{\Delta} CH_3-CH=CH-CHO \quad (But\text{-}2\text{-}enal)}$$

2. Cross-Aldol Condensation

When aldol condensation occurs between two distinct carbonyl species, each possessing α-hydrogens, a statistical mixture of four condensation products is obtained (two self-aldol and two cross-aldol products). For example, reacting ethanal with propanal yields:

- **Self-condensation of ethanal:** But-2-enal ($\mathrm{CH_3-CH=CH-CHO}$)
- **Self-condensation of propanal:** 2-Methylpent-2-enal ($\mathrm{CH_3CH_2-CH=C(CH_3)-CHO}$)
- **Cross-condensation (ethanal electrophile + propanal nucleophile):** 2-Methylbut-2-enal ($\mathrm{CH_3-CH=C(CH_3)-CHO}$)
- **Cross-condensation (propanal electrophile + ethanal nucleophile):** Pent-2-enal ($\mathrm{CH_3CH_2-CH=CH-CHO}$)

When one component lacks an α-hydrogen (e.g., benzaldehyde or formaldehyde) and the other has α-hydrogens (e.g., acetophenone), a single predominant cross-aldol product is obtained (Claisen-Schmidt reaction):

$$\mathrm{C_6H_5CHO + CH_3COC_6H_5 \xrightarrow[293\ K]{OH^-} C_6H_5-CH=CH-CO-C_6H_5 + H_2O \quad (Benzalacetophenone)}$$

3. Cannizzaro Reaction

Aldehydes that **completely lack α-hydrogens** cannot form enolates. Upon heating with concentrated alkali ($50\%\ \mathrm{NaOH}$ or KOH), they undergo intermolecular self-oxidation and reduction (disproportionation). One molecule is reduced to a primary alcohol while another is oxidised to a carboxylate salt:

$$\mathrm{2HCHO + conc.\ KOH \xrightarrow{\Delta} CH_3OH + HCOOK \quad (Methanol + Potassium\ formate)}$$

$$\mathrm{2C_6H_5CHO + conc.\ NaOH \xrightarrow{\Delta} C_6H_5CH_2OH + C_6H_5COONa \quad (Benzyl\ alcohol + Sodium\ benzoate)}$$

> **Crossed Cannizzaro Reaction:** When an aromatic aldehyde reacts with formaldehyde in concentrated alkali, formaldehyde is always preferentially oxidised to formate because nucleophilic attack by hydroxide on the unhindered, highly electrophilic formaldehyde carbon is much faster:
>
> $$\mathrm{C_6H_5CHO + HCHO + conc.\ NaOH \longrightarrow C_6H_5CH_2OH + HCOONa}$$

4. Electrophilic Aromatic Substitution

The carbonyl group is a strong electron-withdrawing group ($-I$ and $-R$), which deactivates the aromatic ring and directs incoming electrophiles to the **meta-position**:

$$\mathrm{C_6H_5CHO + HNO_3 \xrightarrow[273\text{--}283\ K]{H_2SO_4} \textit{m}\text{-}O_2N-C_6H_4-CHO \quad (\textit{m}\text{-}Nitrobenzaldehyde)}$$

7. Carboxylic Acids: Structure, Acidity, and Reactions

7.1 Structure of the Carboxyl Group

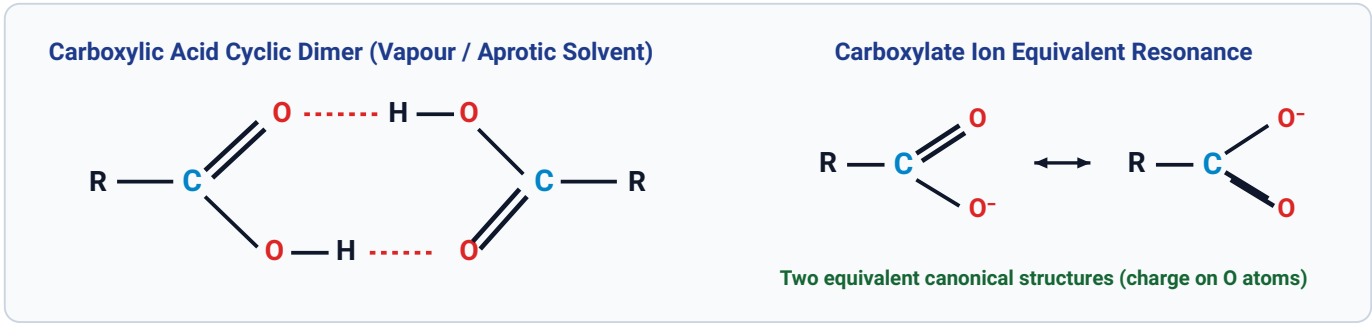
The carboxyl carbon is sp^2-hybridised with planar bond angles of $\sim 120^\circ$. The carboxylic carbon is substantially **less electrophilic** than an aldehydic or ketonic carbonyl carbon because the lone pair of electrons on the hydroxyl oxygen is delocalised into the carbonyl π-system:

$$\mathrm{-C(=O)-\ddot{O}H \longleftrightarrow -C(O^-)=\overset{+}{O}H}$$

7.2 Physical State & Intermolecular Hydrogen Bonding

Carboxylic acids exhibit extensive intermolecular hydrogen bonding. In the liquid and solid states, and even in the vapour phase and non-polar solvents (benzene), they associate into stable **eight-membered cyclic dimers**. Consequently, their boiling points are higher than those of alcohols of equivalent molecular mass.

BOSONANCE



Tutorial Diagram 3: Cyclic eight-membered carboxylic acid dimer and resonance stabilisation of the carboxylate anion.

7.3 Acidity and Substituent Effects

Carboxylic acids are substantially more acidic than alcohols ($pK_a \approx 16$) and phenols ($pK_a \approx 10$). While phenoxide has more resonance contributors, its negative charge is placed on carbon atoms of lower electronegativity. In contrast, the carboxylate anion distributes its negative charge equally across **two highly electronegative oxygen atoms**, creating extraordinary resonance stabilisation:

$$\mathrm{R-COOH + H_2O \rightleftharpoons R-COO^- + H_3O^+} \quad ; \quad K_a = \frac{[\mathrm{RCOO^-}][\mathrm{H_3O^+}]}{[\mathrm{RCOOH}]} \quad ; \quad pK_a = -\log K_a$$

Lower pK_a values denote stronger acids:

- **Electron-Withdrawing Groups (EWGs)** ($-\mathrm{NO_2}$, $-\mathrm{CN}$, $-\mathrm{F}$, $-\mathrm{Cl}$) disperse negative charge by $-I$ and $-R$ effects, stabilising the carboxylate anion and increasing acidity.
- **Electron-Donating Groups (EDGs)** ($-\mathrm{CH_3}$, $-\mathrm{OCH_3}$) intensify negative charge on the carboxylate group, destabilising it and lowering acidity.

Relative Acid-Strengthening Power of Common Groups:

$$-\mathrm{Ph} < -\mathrm{I} < -\mathrm{Br} < -\mathrm{Cl} < -\mathrm{F} < -\mathrm{CN} < -\mathrm{NO_2} < -\mathrm{CF_3}$$

Benchmark Acidity Hierarchy:

$$\mathrm{CF_3COOH} > \mathrm{CCl_3COOH} > \mathrm{CHCl_2COOH} > \mathrm{NO_2CH_2COOH} > \mathrm{NC-CH_2COOH} > \mathrm{FCH_2COOH} > \mathrm{ClCH_2COOH} > \mathrm{BrCH_2COOH} > \mathrm{HCOOH} > \mathrm{ClCH_2CH_2COOH} > \mathrm{C_6H_5COOH} > \mathrm{C_6H_5CH_2COOH} > \mathrm{CH_3COOH} > \mathrm{CH_3CH_2COOH}$$

Substituted Benzoic Acids: Direct attachment of a phenyl group increases acidity relative to acetic acid because the sp^2-hybridised ring carbon is more electronegative than an sp^3 alkyl carbon. Ring substituents alter this value predictably:

- 4-Nitrobenzoic acid ($pK_a = 3.41$) > Benzoic acid ($pK_a = 4.19$) > 4-Methoxybenzoic acid ($pK_a = 4.46$)

7.4 Chemical Reactions of Carboxylic Acids

1. **Reactions Involving $\mathrm{O-H}$ Cleavage:** Reaction with electropositive metals (Na) liberates hydrogen gas ($\mathrm{H_2}$). Unlike phenols, carboxylic acids react with weaker bases like aqueous sodium bicarbonate ($\mathrm{NaHCO_3}$) or sodium carbonate to evolve brisk bubbles of carbon dioxide ($\mathrm{CO_2}$):
$$\mathrm{R-COOH + NaHCO_3 \longrightarrow R-COONa + H_2O + CO_2\uparrow} \quad (\text{Diagnostic test for } -\mathrm{COOH})$$
2. **Acid Anhydride Formation:** Heating carboxylic acids with a dehydrating mineral acid ($\mathrm{H_2SO_4}$) or phosphorus pentoxide ($\mathrm{P_2O_5}$) yields an acid anhydride:
$$\mathrm{2CH_3COOH \xrightarrow[\Delta]{P_2O_5} CH_3-CO-O-CO-CH_3 + H_2O}$$
3. **Fischer Esterification:** Reversible condensation with alcohols in the presence of an acid catalyst ($\mathrm{conc.\ H_2SO_4}$ or dry HCl gas). Water or ester is distilled off continuously to drive the equilibrium forward:
$$\mathrm{R-COOH + R'OH \xrightleftharpoons{H^+} R-COOR' + H_2O}$$
Mechanism Note: Nucleophilic acyl substitution; the $-\mathrm{OH}$ group is cleaved from the acid, while the $-\mathrm{H}$ comes from the alcohol.
4. **Reactions with $\mathrm{PCl_5}$, $\mathrm{PCl_3}$, and $\mathrm{SOCl_2}$:** Converts carboxylic acids to acyl chlorides:
$$\mathrm{RCOOH + PCl_5 \longrightarrow RCOCl + POCl_3 + HCl}$$
$$\mathrm{3RCOOH + PCl_3 \longrightarrow 3RCOCl + H_3PO_3}$$
$$\mathrm{RCOOH + SOCl_2 \longrightarrow RCOCl + SO_2\uparrow + HCl\uparrow}$$
NEET Point: Thionyl chloride ($\mathrm{SOCl_2}$) is preferred because both by-products ($\mathrm{SO_2}$ and HCl) are volatile gases, leaving the pure acyl chloride behind.
5. **Reaction with Ammonia:** Forms an ammonium salt which eliminates water on thermal heating to form an amide:
$$\mathrm{CH_3COOH + NH_3 \rightleftharpoons CH_3COONH_4 \xrightarrow[-H_2O]{\Delta} CH_3CONH_2} \quad (\text{Ethanamide})$$
Phthalic acid forms ammonium phthalate $\xrightarrow[-2H_2O]{\Delta}$ phthalamide $\xrightarrow[-NH_3]{\text{strong }\Delta}$ **phthalimide**.

BOSONANCE

6. **Selective Reduction to Primary Alcohols:** Carboxylic acids are reduced directly to primary alcohols by lithium aluminium hydride ($\mathrm{LiAlH_4}$) or diborane ($\mathrm{B_2H_6}$):

$$\mathrm{R-COOH} \xrightarrow[\text{2. } \mathrm{H_3O^+}]{\text{1. } \mathrm{B_2H_6} \text{ in ether}} \mathrm{R-CH_2OH}$$

NEET Traps: Sodium borohydride ($\mathrm{NaBH_4}$) **does NOT reduce** carboxylic acids. Diborane selectively reduces carboxyl groups without reducing esters, nitro, or halo substituents.

7. **Decarboxylation:** Heating sodium carboxylates with soda-lime (mixture of NaOH and CaO in a 3:1 mass ratio) produces an alkane with one fewer carbon:

$$\mathrm{R-COONa} \xrightarrow[\Delta]{\mathrm{NaOH+CaO}} \mathrm{R-H} + \mathrm{Na_2CO_3}$$

8. **Hell-Volhard-Zelinsky (HVZ) Reaction:** Carboxylic acids possessing α-hydrogens are halogenated at the α-position by chlorine or bromine in the presence of red phosphorus:

$$\mathrm{R-CH_2-COOH} \xrightarrow[\text{2. } \mathrm{H_2O}]{\text{1. } \mathrm{X_2/red\ P}} \mathrm{R-CH(X)-COOH} \quad (\mathrm{X = Cl,\ Br})$$

9. **Ring Substitution of Aromatic Acids:** The carboxyl group is strongly deactivating and meta-directing. It undergoes electrophilic halogenation and nitration:

$$\mathrm{C_6H_5COOH + Br_2} \xrightarrow{\mathrm{FeBr_3}} m\text{-}\mathrm{Br-C_6H_4-COOH}$$

> **NEET Mandatory Check:** Benzoic acid and aromatic carboxylic acids **DO NOT undergo Friedel-Crafts alkylation or acylation**. The $-COOH$ group is strongly electron-withdrawing, and the Lewis acid catalyst ($\mathrm{AlCl_3}$) coordinates strongly with the carboxyl oxygen, completely poisoning catalytic activity.

8. Step-by-Step Worked Conversion Problems

Worked Example 1: Multi-Step Structural Deduction

An organic compound **A** ($\mathrm{C_8H_8O}$) forms an orange-red precipitate with 2,4-DNP reagent and gives a yellow precipitate on heating with $\mathrm{I_2}$ and NaOH. It neither reduces Tollens' reagent nor decolourises bromine water. Vigorous oxidation of **A** with chromic acid yields compound **B** ($\mathrm{C_7H_6O_2}$). Identify **A** and **B** and rationalise the observations.

Step-by-Step Analysis:

1. Reaction with 2,4-DNP confirms the presence of an aldehyde or ketone carbonyl group.
2. Failure to reduce Tollens' reagent excludes aldehydes; compound **A** must be a ketone.
3. Positive iodoform test ($\mathrm{I_2 + NaOH}$) confirms a methyl ketone fragment: $\mathrm{-CO-CH_3}$.
4. Subtracting $\mathrm{-COCH_3}$ (molecular formula $\mathrm{C_2H_3O}$) from $\mathrm{C_8H_8O}$ leaves $\mathrm{C_6H_5}$ (a monosubstituted phenyl ring).
5. The structure of **A** is **acetophenone** ($\mathrm{C_6H_5COCH_3}$).
6. Vigorous oxidation of acetophenone with chromic acid oxidises the acetyl side-chain completely to form benzoic acid:

$$\mathrm{C_6H_5COCH_3} \xrightarrow{\mathrm{H_2CrO_4}} \mathrm{C_6H_5COOH\ (\mathbf{B})}$$

Final Identification: **A** is acetophenone (1-phenylethan-1-one); **B** is benzoic acid (benzenecarboxylic acid).

Worked Example 2: Distinguishing Pairs of Functional Isomers

Detail single laboratory test-tube reactions to differentiate between the following pairs:
(a) Pentan-2-one and Pentan-3-one
(b) Benzaldehyde and Acetophenone
(c) Benzoic acid and Phenol

Analytical Resolutions:

- **(a) Pentan-2-one vs Pentan-3-one:** Add $\mathrm{I_2}$ and NaOH solution and warm gently. Pentan-2-one contains a methyl ketone group ($\mathrm{CH_3COCH_2CH_2CH_3}$) and forms a yellow precipitate of iodoform ($\mathrm{CHI_3}$). Pentan-3-one lacks a $\mathrm{-COCH_3}$ unit and yields no precipitate.
- **(b) Benzaldehyde vs Acetophenone:** Warm with Tollens' reagent ($\mathrm{[Ag(NH_3)_2]^+}$). Benzaldehyde oxidises to benzoate and deposits a bright silver mirror on the glass wall. Acetophenone, being a ketone, gives no reaction. (Alternatively, acetophenone gives a positive iodoform test, whereas benzaldehyde is negative).
- **(c) Benzoic acid vs Phenol:** Add a saturated aqueous solution of sodium bicarbonate ($\mathrm{NaHCO_3}$). Benzoic acid is sufficiently acidic to liberate $\mathrm{CO_2}$ with brisk effervescence. Phenol is too weakly acidic ($pK_a \approx 10$) to decompose bicarbonate and shows no gas evolution.

BOSONANCE

9. NEET UG Exam-Aligned Practice Assessment

NEET STANDARD MCQ **Practice Question 1**

Arrange the following compounds in order of decreasing reactivity toward nucleophilic addition reactions:
(I) Ethanal **(II)** Propanal **(III)** Propanone **(IV)** Butanone

(A) I > II > III > IV

(B) IV > III > II > I

(C) II > I > IV > III

(D) I > III > II > IV

Correct Answer: (A)
Explanation: Aldehydes are more reactive than ketones due to smaller steric crowding and a weaker $+I$ inductive effect (one alkyl group vs. two). Comparing the two aldehydes: ethanal has one small methyl group, whereas propanal has a larger ethyl group; thus, ethanal is more reactive than propanal. Comparing the two ketones: propanone has two methyl groups, while butanone has a bulkier ethyl group; thus, propanone is more reactive than butanone. The correct decreasing order is Ethanal (I) > Propanal (II) > Propanone (III) > Butanone (IV).

NEET STANDARD MCQ **Practice Question 2**

In the Rosenmund reduction, barium sulphate ($\mathrm{BaSO_4}$) added along with the palladium catalyst acts as:

(A) A promoter to accelerate the reduction

(B) A catalyst poison to prevent over-reduction to primary alcohol

(C) An oxidising agent

(D) An emulsifier to increase surface area

Correct Answer: (B)
Explanation: Palladium alone is an active hydrogenation catalyst that would rapidly reduce the newly formed aldehyde to a primary alcohol. Partially poisoning the catalyst with $\mathrm{BaSO_4}$ (often in the presence of quinoline or sulfur) moderates catalytic activity, stopping the reaction cleanly at the aldehyde stage.

NEET STANDARD MCQ **Practice Question 3**

Which of the following carbonyl compounds does NOT yield an α,β-unsaturated aldehyde or ketone when treated with dilute sodium hydroxide?

(A) Ethanal

(B) 2-Methylpropanal

(C) 2,2-Dimethylpropanal

(D) Cyclohexanone

Correct Answer: (C)
Explanation: Aldol condensation requires at least one acidic α-hydrogen. 2,2-Dimethylpropanal, $(\mathrm{CH_3})_3\mathrm{C}-\mathrm{CHO}$, has no hydrogen atoms on its α-carbon. Under basic conditions, it cannot form an enolate and instead undergoes the Cannizzaro disproportionation reaction.

ASSERTION-REASON **Practice Question 4**

Assertion (A): Benzaldehyde does not reduce Fehling's solution.
Reason (R): In aromatic aldehydes, the carbonyl carbon is resonance-stabilised by delocalisation of ring π-electrons, rendering it a weaker reducing agent than aliphatic aldehydes.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Correct Answer: (A)
Explanation: Fehling's reagent is a milder oxidising agent than Tollens' reagent. Because the carbonyl group of benzaldehyde is conjugated with the benzene ring, the $\mathrm{C}-\mathrm{H}$ bond of the formyl group has higher bond dissociation energy and the carbonyl carbon is less electrophilic. As a result, aromatic aldehydes reduce ammoniacal silver nitrate (Tollens'), but fail to reduce copper(II) tartrate (Fehling's).

BOSONANCE

NEET STANDARD MCQ **Practice Question 5**

Identify the correct sequence of increasing acid strength (pK_a value decreasing):

(A) $\mathrm{CH_3COOH < HCOOH < ClCH_2COOH < FCH_2COOH}$

(B) $\mathrm{FCH_2COOH < ClCH_2COOH < HCOOH < CH_3COOH}$

(C) $\mathrm{HCOOH < CH_3COOH < ClCH_2COOH < FCH_2COOH}$

(D) $\mathrm{CH_3COOH < ClCH_2COOH < FCH_2COOH < HCOOH}$

Correct Answer: (A)
Explanation: Electron-withdrawing inductive effects ($-I$) stabilise the carboxylate anion and enhance acidity: $\mathrm{F} > \mathrm{Cl}$. In acetic acid ($\mathrm{CH_3COOH}$), the methyl group exerts an electron-donating inductive effect ($+I$), destabilising the carboxylate anion and making it weaker than formic acid (HCOOH). The correct order of increasing acidity is $\mathrm{CH_3COOH < HCOOH < ClCH_2COOH < FCH_2COOH}$.

NEET STANDARD MCQ **Practice Question 6**

An organic compound with molecular formula $\mathrm{C_9H_{10}O}$ forms a 2,4-DNP derivative, reduces Tollens' reagent, and undergoes the Cannizzaro reaction. Upon vigorous oxidation with alkaline $\mathrm{KMnO_4}$, it yields benzene-1,2-dicarboxylic acid. The compound is:

(A) 2-Ethylbenzaldehyde

(B) 3-Ethylbenzaldehyde

(C) 4-Methylacetophenone

(D) 2,4-Dimethylbenzaldehyde

Correct Answer: (A)
Explanation:

1. Forms 2,4-DNP and reduces Tollens' reagent $\Longrightarrow$ contains an aldehyde group ($-\mathrm{CHO}$).
2. Undergoes the Cannizzaro reaction $\Longrightarrow$ the $-\mathrm{CHO}$ group is directly attached to the benzene ring (no α-H).
3. Vigorous oxidation produces benzene-1,2-dicarboxylic acid (phthalic acid) $\Longrightarrow$ the two side-chains must be ortho (1,2-relationship) on the benzene ring.
4. Formula is $\mathrm{C_9H_{10}O}$. The benzene ring uses 6 carbons, the aldehyde uses 1 carbon, leaving an ethyl group ($-\mathrm{C_2H_5}$) at position 2.

The compound is **2-ethylbenzaldehyde**.

MATCHING TYPE **Practice Question 7**

Match the chemical transformations in **List I** with their appropriate reagents in **List II**:

List I (Transformation)	List II (Reagent)
(P) Toluene $\longrightarrow$ Benzaldehyde	(1) $\mathrm{Br_2/Red\ P}$ followed by $\mathrm{H_2O}$
(Q) Benzoyl chloride $\longrightarrow$ Benzaldehyde	(2) $\mathrm{CrO_2Cl_2}$ in $\mathrm{CS_2}$ followed by $\mathrm{H_3O^+}$
(R) Acetic acid $\longrightarrow$ Bromoacetic acid	(3) $\mathrm{H_2,\ Pd\text{-}BaSO_4}$
(S) Propanone $\longrightarrow$ Propane	(4) $\mathrm{Zn\text{-}Hg/conc.\ HCl}$

(A) P-2, Q-3, R-1, S-4

(B) P-3, Q-2, R-1, S-4

(C) P-2, Q-3, R-4, S-1

(D) P-4, Q-1, R-2, S-3

Correct Answer: (A)
Explanation: (P) Toluene $\rightarrow$ Benzaldehyde via chromyl chloride is the Etard reaction (2).
(Q) Benzoyl chloride $\rightarrow$ Benzaldehyde via $\mathrm{H_2/Pd - BaSO_4}$ is the Rosenmund reduction (3).
(R) Acetic acid $\rightarrow$ Bromoacetic acid via $\mathrm{Br_2/Red\ P}$ is the Hell-Volhard-Zelinsky (HVZ) reaction (1).
(S) Propanone $\rightarrow$ Propane using $\mathrm{Zn - Hg/conc.\ HCl}$ is the Clemmensen reduction (4).

BOSONANCE

NEET STANDARD MCQ **Practice Question 8**

Benzoic acid does NOT undergo Friedel-Crafts reaction because:

(A) The $-\mathrm{COOH}$ group is an ortho/para director

(B) The $-\mathrm{COOH}$ group is strongly deactivating and coordinates with the Lewis acid catalyst $\mathrm{AlCl_3}$

(C) Benzoic acid is unstable at room temperature

(D) The benzene ring becomes excessively nucleophilic

Correct Answer: (B)
Explanation: The carboxyl group ($-\mathrm{COOH}$) is strongly meta-directing and deactivates the aromatic ring toward electrophilic attack. Furthermore, the Lewis acid catalyst ($\mathrm{AlCl_3}$) forms a coordination complex with the lone pairs on the carboxyl oxygen, completely deactivating both the catalyst and the ring.

NEET STANDARD MCQ **Practice Question 9**

Which of the following statements regarding the haloform reaction is INCORRECT?

(A) Acetone gives a positive iodoform test with $\mathrm{I_2/NaOH}$

(B) Pentan-3-one gives a yellow precipitate of iodoform

(C) Ethanol responds positively to the iodoform reaction

(D) Carbon-carbon double bonds present in the substrate remain unaffected during sodium hypohalite oxidation

Correct Answer: (B)
Explanation: Pentan-3-one ($\mathrm{CH_3CH_2-CO-CH_2CH_3}$) is a symmetrical diethyl ketone. It does not contain a terminal methyl ketone group ($\mathrm{CH_3-C=O}$) and cannot undergo base-catalysed polyhalogenation at a methyl carbon; hence, it does not give the iodoform test. Ethanol responds because it oxidises in situ to acetaldehyde ($\mathrm{CH_3CHO}$).

ASSERTION-REASON **Practice Question 10**

Assertion (A): Carboxylic acids do not form addition products with hydroxylamine or semicarbazide like aldehydes and ketones.
Reason (R): The electrophilicity of the carboxyl carbonyl carbon is significantly reduced by resonance involving the lone pair on the hydroxyl oxygen atom.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Correct Answer: (A)
Explanation: In carboxylic acids, resonance donation of lone pairs from the $-\mathrm{OH}$ oxygen into the carbonyl π^*-orbital disperses the partial positive charge on carbon:

$$-\mathrm{C(=O)-OH} \longleftrightarrow -\mathrm{C(O^-)=\overset{+}{O}H}$$

As a result, the carbonyl carbon is far less electrophilic than that in aldehydes or ketones, preventing nucleophilic addition by weak bases such as $\mathrm{NH_2OH}$ or $\mathrm{NH_2NHCONH_2}$.

10. Common Pitfalls & High-Yield Revision Checklist

Topic / Reaction	Common Student Error	Accurate NEET Chemical Rule
Fehling's Test	Assuming benzaldehyde gives a positive Fehling's test because it is an aldehyde.	**Aromatic aldehydes do not reduce Fehling's solution.** Benzaldehyde reduces Tollens' reagent, but not Fehling's.
Semicarbazide Attack	Depicting nucleophilic attack from the $-\mathrm{CONH_2}$ terminus.	The amide amino group is resonance-deactivated by the adjacent $>\mathrm{C=O}$. Nucleophilic attack occurs exclusively via the **hydrazine** $-\mathrm{NH_2}$ **group**.
Crossed Cannizzaro	Assuming benzaldehyde oxidises to benzoate when mixed with formaldehyde in concentrated alkali.	**Formaldehyde is always oxidised to formate** ($\mathrm{HCOO^-}$) because it is less hindered and more electrophilic, while benzaldehyde is reduced to benzyl alcohol.
Clemmensen vs Wolff-Kishner	Using Clemmensen reduction on keto-compounds containing acid-sensitive groups (e.g., $-\mathrm{OH}$, acetal).	Clemmensen uses concentrated HCl (acidic conditions). For acid-sensitive substrates, use **Wolff-Kishner** ($\mathrm{NH_2NH_2/KOH}$, basic conditions).
Carboxylic Acid Reduction	Writing $\mathrm{NaBH_4}$ as a reagent to reduce $-\mathrm{COOH}$ to primary alcohols.	$\mathrm{NaBH_4}$ **cannot reduce carboxylic acids or esters**. Use $\mathrm{LiAlH_4}$ or diborane ($\mathrm{B_2H_6}$).

Topic / Reaction	Common Student Error	Accurate NEET Chemical Rule
Hell-Volhard-Zelinsky (HVZ)	Attempting HVZ on formic acid or 2,2-dimethylpropanoic acid.	The HVZ reaction strictly requires **at least one α-hydrogen**. Formic acid and pivalic acid do not react.

11. Chapter Summary

- **Carbonyl Geometry:** Planar sp^2 carbon with $\sim 120^\circ$ bond angles. Polarity ($\mathrm{C}^{\delta+} = \mathrm{O}^{\delta-}$) makes the carbon electrophilic.
- **Nucleophilic Addition:** Aldehydes react faster than ketones due to steric freedom and minimal $+I$ inductive shielding. Benzaldehyde is less reactive than propanal due to resonance delocalisation from the phenyl ring.
- **Key Preparations:**
  - Acyl chloride $\rightarrow$ Aldehyde: Rosenmund reduction ($\mathrm{H_2/Pd - BaSO_4}$).
  - Nitrile $\rightarrow$ Aldehyde: Stephen reduction ($\mathrm{SnCl_2 + HCl}$) or DIBAL-H at $195\,\mathrm{K}$.
  - Methylbenzene $\rightarrow$ Benzaldehyde: Etard reaction ($\mathrm{CrO_2Cl_2}$) or $\mathrm{CrO_3/(CH_3CO)_2O}$.
  - Benzene formylation: Gatterman-Koch reaction ($\mathrm{CO + HCl/anh.\ AlCl_3, CuCl}$).
- **Distinguishing Tests:** Tollens' oxidises all aldehydes. Fehling's oxidises only aliphatic aldehydes. The iodoform test identifies $\mathrm{CH_3 - C = O}$ and $\mathrm{CH_3 - CH(OH) -}$ groups.
- **α-Hydrogen Reactions:** Dilute alkali gives aldol condensation (requires ≥ 1 α-H). Concentrated alkali causes Cannizzaro disproportionation (requires no α-H).
- **Carboxylic Acid Acidity:** More acidic than phenols because the carboxylate charge is delocalised over two equivalent oxygen atoms. EWGs increase acidity; EDGs decrease acidity.
- **Carboxylic Acid Transformations:** Cleavage of $-\mathrm{OH}$ gives anhydrides, esters, and acyl chlorides ($\mathrm{SOCl_2}$ preferred). α-Halogenation occurs via the HVZ reaction. Benzoic acid does not undergo Friedel-Crafts reactions.

BOSONANCE